P510/2 PHYSICS

Paper 2

Time: 3 Hours

INSTRUCTIONS TO THE CANDIDATES:

- ✓ *This examination paper consists of **two** compulsory items*
- ✓ *Use **blue or blue or black** ink. Any work done in pencil except graphs and diagrams will not be scored*
- ✓ *Scores are awarded mainly for clear record of observations actually made ,for their suitability, accuracy and for the use made of them.*
- ✓ *Details on question paper should not be repeated in the response.*
- ✓ *Graph paper(s) is or are provided*

Item one.

A TV broadcasting station serving viewers in a hilly region is experiencing poor performance in a dish-type reception system used to collect and focus television signals for people watching World Cup games. The system, which had previously functioned effectively, now produces weak, inconsistent, and distorted signals,resulting in poor picture and sound quality during broadcasts.

Investigations revealed that the original dish reflector, which had the required **focal length within the range of 80 cm to 120 cm** for efficient signal focusing,was stolen and replaced with a substitute dish of unknown specifications. Since the replacement, the performance of the system has deteriorated considerably.

A technical expert familiar with the installation stated that the focal length of the replacement dish is ten times that of an equivalent concave mirror available in the laboratory. However, due to the difficulties and costs involved in dismantling the installed dish for direct testing, the expert provided the concave mirror so that the focal length of the dish could be determined indirectly.

Although the expert suspects that the replacement dish may not possess the required focal length for effective focusing of TV signals, he could not rule out the possibility that the poor performance may be due to other factors within the reception system. He therefore recommended a scientific investigation to establish the actual causeof the problem.

Task

As an AdIvanced Level Physics student, carry out scientific investigations using at least two different methods to ascertain whether the poor performance of the TV

reception system is due to the replacement dish having an incorrect focal length or otherwise justify your conclusion.

Item two.

A laboratory technician procured rolls of constantan wire for use in a Physics practical examination. The wire required for the examination should possess a **resistance per metre of 4.5** Ω m-1. However, when the supplier delivered the rolls of bare wire, none of them were labelled, making it necessary to verify whether they met the required specification. To establish their suitability, the technician designed an investigation using two separate methods. In the first method, he planned to determine the resistance of the wire using an ammeter, a voltmeter, a metre rule, and connecting leads. In the second method, he intended to use a metre bridge together with a known standard resistor, R, a metre rule, and connecting wires to obtain more reliable results.

Unfortunately, before carrying out the investigation, the technician experienced bereavement and had to travel to attend the burial of a relative. As a result, the Director of Studies (DOS) has assigned you the responsibility of testing one of the wires supplied and availed you with steps to follow.

Task

Use the availed apparatus and instructions to carry out a scientific investigation to appropriately advise the DOS.

Method I

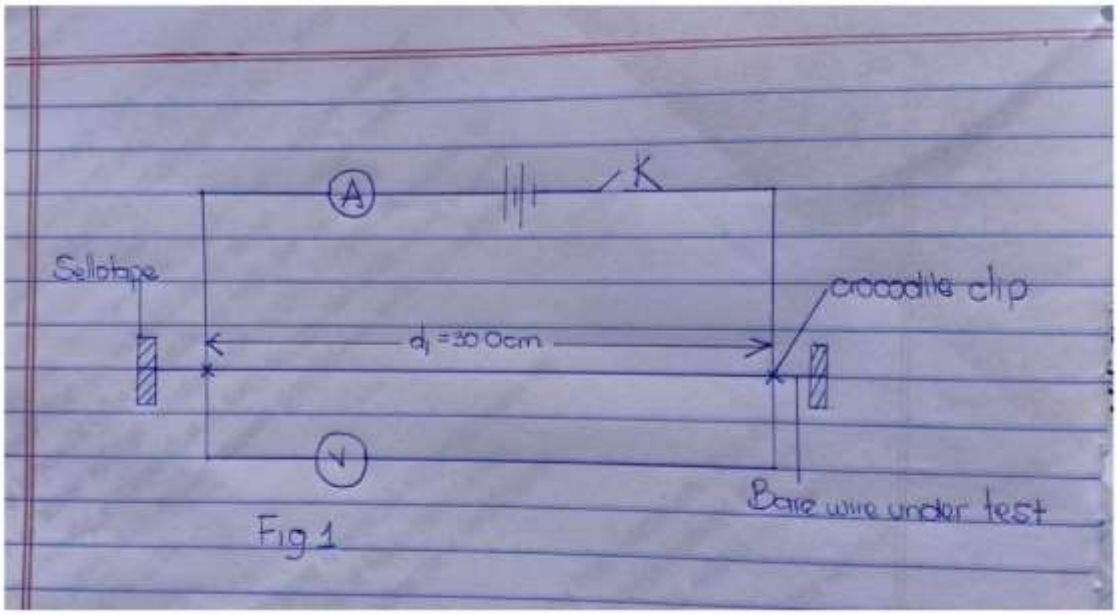


Fig. 1

(a) Connect the circuit as shown in the figure 1.

(b) Adjust the distanced_1, between the crocodile clips to 30.0cm

(c) Close switch K

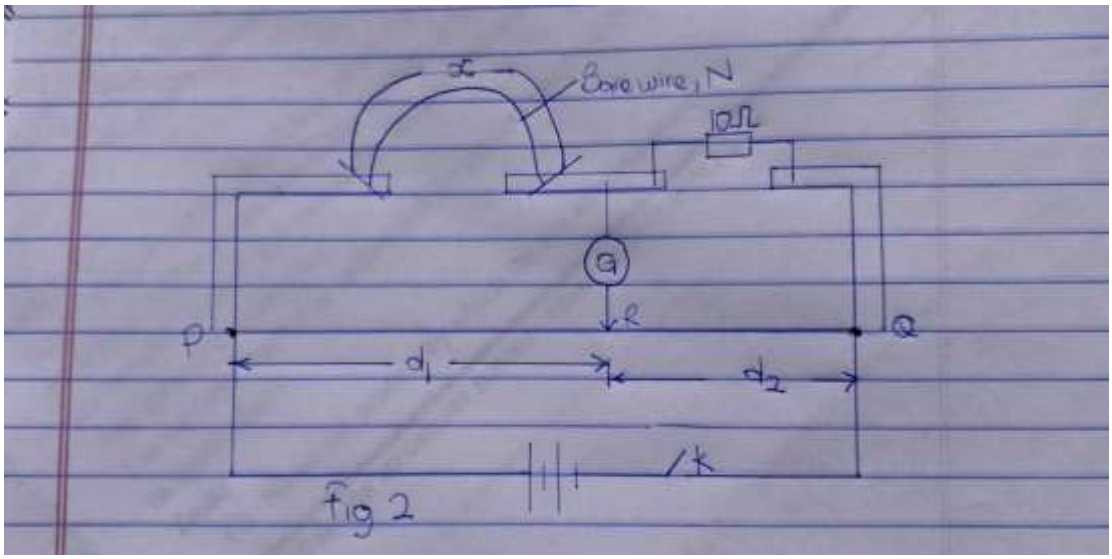
(d) Read and record the ammeter reading,$I_1$and voltmeter reading,V_1,

(e) Open the switch.

(f) Repeat steps (b) to (d) for$d_2 = 60.0cm$.

(g)Calculate the resistance per metreμ_1, of the bare wire using t he expression: $\mu_1 = \frac{1}{2}(\frac{V_1}{I_1 d_1} + \frac{V_2}{I_2 d_2})$

Method 11



(a) Set up the apparatus as shown in the fig.2.

(b) Adjust the length x, of the bare wire to 0.800 m.

(c) Close the switch K.

(d) Move the jockey J along the metre bridge slide wire PQ until a **null** point is reached.

(e) Read and record, the balance length,d_1

(f) Open the switch K.

(g) Repeat steps (b) to (e) for other values of,x=0.700,0.600,0.500,0.400 and 0.300 m.

(h) Record the results in a table including values of $d_2 = (1 - d_1)$ and $R = 10\frac{d_1}{d_2}$

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In addition to the apparatus in the physics laboratory, the following should be provided to each candidate:

Item 1

- o 1-Concave mirror of focal length 10 cm in holder
- o 1-metre rule
- o 1- white screen with wire gauze
- o 1- white screen
- o A torch bulb in a holder
- o A pair of dry cells in a holder
- o 1-switch
- o 4 connecting wires of about 50 cm long

Item 2

- o 1-ammeter of range 0-1 A
- o 1-voltmeter of range 0-3 A or 0-6 V
- o A pair of dry cells in a holder of each e.m.f of 1.5 V
- o 1- Constantine wire SWG 28 of length about 105 cm, labeled N
- o 1-10 Ω resistor labeled R
- o 1- metre bridge
- o 1- switch
- o 10-connecting wire of length 50 cm
- o 1- metre rule
- o 2- crocodile clips
- o 1-jockey
- o 1-micrometer screw gauge
- o Zero centre galvanometer